CARDIFF UNIVERSITY
EXAMINATION PAPER

Academic Year :	2018-2019
Examination Period :	Spring
Examination Paper Number :	MA 0367
Examination Paper Title :	Time Series Analysis and Forecasting

Do not turn this page over until instructed to do by the Senior Invigilator.

Structure of Examination Paper:

There are Five pages.
There are FOUR questions in total.
There are no appendices.
The mark obtainable for each question is 25.

Students to be provided with:

The following items of stationery are to be provided at the start of the examination:
One answer book.

Instructions to Students:

Answer **THREE** questions

The use of a translation dictionary between English or Welsh and another language, provided that it bears an appropriate departmental stamp, is permitted in this examination.

Question 1

(i) State the Mercer expansion and the Karhunen–Loeve expansion.

[2]

(ii) Explain how to compute the correlation function $\rho(k)$, $k = 1, 2, 3, \ldots$, for an AR(p) process with known coefficients using the Yule-Walker equations.

[3]

(iii) Write R code for simulating and depicting a realization from the Poisson process.

[3]

(iv) Find two values of the covariance function $R(0)$ and $R(1)$ for the ARMA(1,1) process $x_t - 0.5x_{t-1} = \epsilon_t + 0.4\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$.

[3]

(v) Consider the causal ARMA(2,1) process $x_t = x_{t-1} - 0.5x_{t-2} + \epsilon_t + 0.6\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$ of this ARMA process.

[6]

(vi) Suppose that the average precipitation in Barcelona in successive months was approximated by the AR(2) process $x_t - 11 = 0.5(x_{t-1} - 11) - 0.4(x_{t-2} - 11) + \epsilon_t$, where ϵ_t is Gaussian white noise with zero mean and variance 0.7. Suppose that the time series 12, 11, 15, 14 is observed in recent months. Find the predictions for the next two months, the mean square error of predictors, and the confidence intervals with confidence level $1 - \alpha$.

[4]

(vii) Find the covariance function for a discrete weakly stationary process x_t with spectral density $f(\lambda) = 1 + |\lambda|$.

[4]

Question 2

(i) Give the definitions of an ARMA(p, q) process with zero mean and an ARMA(p, q) process with mean μ.

[2]

(ii) State and prove the condition of causality of the AR(2) process in terms of coefficients.

[4]

(iii) Write R code for estimating the parameters of an AR(3) process for the time series: 7, 5, 3, 1, 4, 2, 6, 3 via solving a suitable system of equations.

[4]

(iv) Is the stochastic process with covariance function $R(k) = \frac{9}{(1+|k|^3)^{3/8}}$, $k \in \mathbb{Z}$, short-range or long-range dependent? Explain your answer.

[2]

(v) Consider the time series $x_1 = 52$, $x_2 = 57$, $x_3 = 53$ and assume that the observations correspond to a stationary normal process with covariance function $R(k) = 2/\sqrt{1+k^2}$. Compute the best linear predictor for the next observation $\hat{x}_4$ and also the prediction bounds for x_4 with confidence level $1 - \alpha$.

[6]

(vi) A shop sells books. Sales of books (in units of 100) over the past 12 weeks are $x_1, \ldots, x_{12}$ with $x_{12} = 7.2$. The shop believes that the AR(1) process $x_t - 6 = 0.7(x_{t-1} - 6) + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is Gaussian white noise with zero mean and variance 2.89, is relevant for forecasting. Forecast sales for weeks 13, 14, 15, and 16 and give an expression for the confidence interval of future sales for each of the weeks with confidence level $1 - \alpha$.

[3]

(vii) Find at least three values of parameter c such that the stochastic process x_t satisfying

$$x_t = -4cx_{t-1} + 2cx_{t-2} + 8c^2x_{t-3} + \varepsilon_t, \quad t = 0, \pm 1, \ldots,$$

is non-stationary, where ε_t is a white noise process with $\mathrm{E}[\varepsilon_t] = 0$ and $\mathrm{Var}\,[\varepsilon_t] = \sigma_\varepsilon^2$.

[4]

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Question 3

(i) Give the definition of a periodogram and give three different mathematical formulas for computing a periodogram.

[2]

(ii) Show that the correlation function $\rho(t)$ of a stationary process is positive semidefinite.

[2]

(iii) Write R code for simulating a realization of a Gaussian process with continuous time, mean function $m(t) = 5/(3+|t|)$ and covariance function $R(t,s) = 2e^{-3|t-s|^{1.4}}$ at points $0.3, 0.8, 1.3, 1.7$.

[3]

(iv) Find a difference equation of the ARMA model for which the correlation function $\rho(k)$ is such that $\rho(1) = 1/\sqrt{2}$, $\rho(2) = 1/4$ and $\rho(k) = 0$ for $k = 3, 4, \ldots$.

[5]

(v) Sketch the temporal and spectral representations of the two time series $x_k = \cos(2\pi k/15) + \epsilon_k$ and $y_k = 2\cos(2\pi k/30) + \varepsilon_k$, where ϵ_k and ε_k are white noise. Explain any differences in your sketches.

[3]

(vi) Suppose that the time series $2, -1, 1, -3, 1, 0$ is a realization from an AR(2) process $x_t - a_1 x_{t-1} - a_2 x_{t-2} = \epsilon_t$, where ϵ_t is white noise. Find estimates of the parameters of the AR(2) process, that is, the coefficients a_1, a_2, and the noise variance $\sigma^2 = \mathrm{Var}(\epsilon_t)$.

[5]

(vii) Consider the stochastic process x_t defined by

$$x_t = \xi \sin(\lambda t + \eta),\ t = 0, \pm 1, \ldots,$$

where ξ is a random variable with zero mean and unit variance, η is a random variable with a uniform distribution on the interval $[-\pi, \pi]$ independent of ξ, and λ is a real number. Show that this process is stationary in the wide sense. Find the covariance function of the process x_t.

[5]

Question 4

(i) Give the definition of the n-order stationary process and the definition of the Wiener process.

[2]

(ii) Let x_t be a continuous stationary process with zero mean and covariance function $R(t)$. Suppose that this process was observed at points $t_1, t_2, \ldots, t_n$, where $t_i < t_{i+1}$. Derive the linear predictor of this process at point t with the smallest mean square error (MSE), $t > t_n$.

[5]

(iii) Write R code for simulating and plotting a realization of length 140 from the ARMA(3,1) process defined by the difference equation $x_t - 0.7x_{t-1} + 0.3x_{t-2} - 0.1x_{t-3} = \epsilon_t + 0.4\epsilon_{t-1}$ with $x_1 = 2.1$, $x_2 = -0.2$ and $x_3 = 0.7$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.5$.

[3]

(iv) Let a process x_t satisfy $x_t + \frac{11}{3}x_{t-1} - \frac{3}{4}x_{t-2} = \epsilon_t - \frac{3}{8}\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is causal or not. Give a reason for your answer.

[3]

(v) Consider the time series $x_k = \sin(9\pi k/28) + 0.5\sin(5\pi k/16) + 2\sin(3\pi k/11)$, $k \in \mathbb{Z}$, and the filter with spectrum $f(\lambda) = 1_{[0.3\pi, 0.35\pi]}(\lambda)$, $\lambda \in (0, \pi)$. Find the filtered series and its spectrum.

[3]

(vi) Suppose that the time series $3, 1, -2, 0, 1, -3$ is a realization from an invertible MA(1) process $x_t = \epsilon_t + b_1\epsilon_{t-1}$, where ϵ_t is white noise. Estimate the parameters of this process: the coefficient b_1 and the noise variance $\sigma^2 = \mathrm{Var}(\epsilon_t)$.

[4]

(vii) Find the variance and the values of the correlation function $\rho(1), \rho(2), \rho(3), \rho(4)$ of the AR(2) process $x_t = 0.8x_{t-1} - 0.5x_{t-2} + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.4$, using Yule-Walker equations.

[5]